

MSMS 408 : Practical 17

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⊕ Question

- (i) Generate a sample of size 1000 from a mixture of Gaussians given by the following PDF.

$$f(x) = 0.3 \cdot \mathcal{N}(-4, 1) + 0.3 \cdot \mathcal{N}(0, 1) + 0.4 \cdot \mathcal{N}(4, 1).$$

Each x_i is generated from $\mathcal{N}(-4, 1)$ with probability 0.3 or from $\mathcal{N}(0, 1)$ with probability 0.3 or from $\mathcal{N}(4, 1)$ with probability 0.4. Assume that you are given only x_i 's and you don't know which of the two Gaussians each x_i comes from.

- (ii) Use Expectation-Maximization algorithm to estimate the mixing proportions, ϕ_1, ϕ_2, ϕ_3 , say; the Gaussian means, μ_1, μ_2, μ_3 , say and the Gaussian variances $\sigma_1^2, \sigma_2^2, \sigma_3^2$ say.
- (iii) Using the parameter estimates, predict which Gaussian each x_i might have been generated from.

⊕ Theory

1

Data simulation :

- Generate $u_1, \dots, u_n \sim U(0, 1)$.
- Set

$$\forall i = 1(1)n, x_i = \begin{cases} x \sim \mathcal{N}(-4, 1) & \text{if } u_i \leq 0.3 \\ x \sim \mathcal{N}(0, 1) & \text{if } 0.3 < u_i \leq 0.6 \\ x \sim \mathcal{N}(4, 1) & \text{if } u_i > 0.6 \end{cases} .$$

2 EM Algorithm :

- $\theta = \{\phi_1, \phi_2, \phi_3, \mu_1, \mu_2, \mu_3, \sigma_1^2, \sigma_2^2, \sigma_3^2\}$.
- Start with an initial value $\theta^{(0)}$.
- Repeat until convergence {

(E-Step) For each i, j :

$$w_j^{(i)} := p\left(z_j^{(i)} = 1 \mid x^{(i)}, \theta^{(t)}\right)$$

(M-Step) Update :

$$\phi_j := \frac{1}{n} \sum_{i=1}^n w_j^{(i)},$$

$$\mu_j := \frac{\sum_{i=1}^n w_j^{(i)} x^{(i)}}{\sum_{i=1}^n w_j^{(i)}},$$

$$\sigma_j^2 := \frac{\sum_{i=1}^n w_j^{(i)} (x^{(i)} - \mu_j)^2}{\sum_{i=1}^n w_j^{(i)}}$$

}.

- 3 Predict $x^{(i)} \in k$ -th Gaussian where $k = \underset{j}{\operatorname{argmax}} w_j^{(i)}$, evaluated on the final estimates of θ .

➔ R Program

```
1 n <- 1000; u <- runif(n)
```

```
theta.true <- c(0.3, 0.3, 0.4, -4, 0, 4, 1, 1, 1)
```

```
phi.true <- theta.true[1:3]
```

```
mu.true <- theta.true[4:6]
```

```
sigma.sq.true <- theta.true[7:9]
```

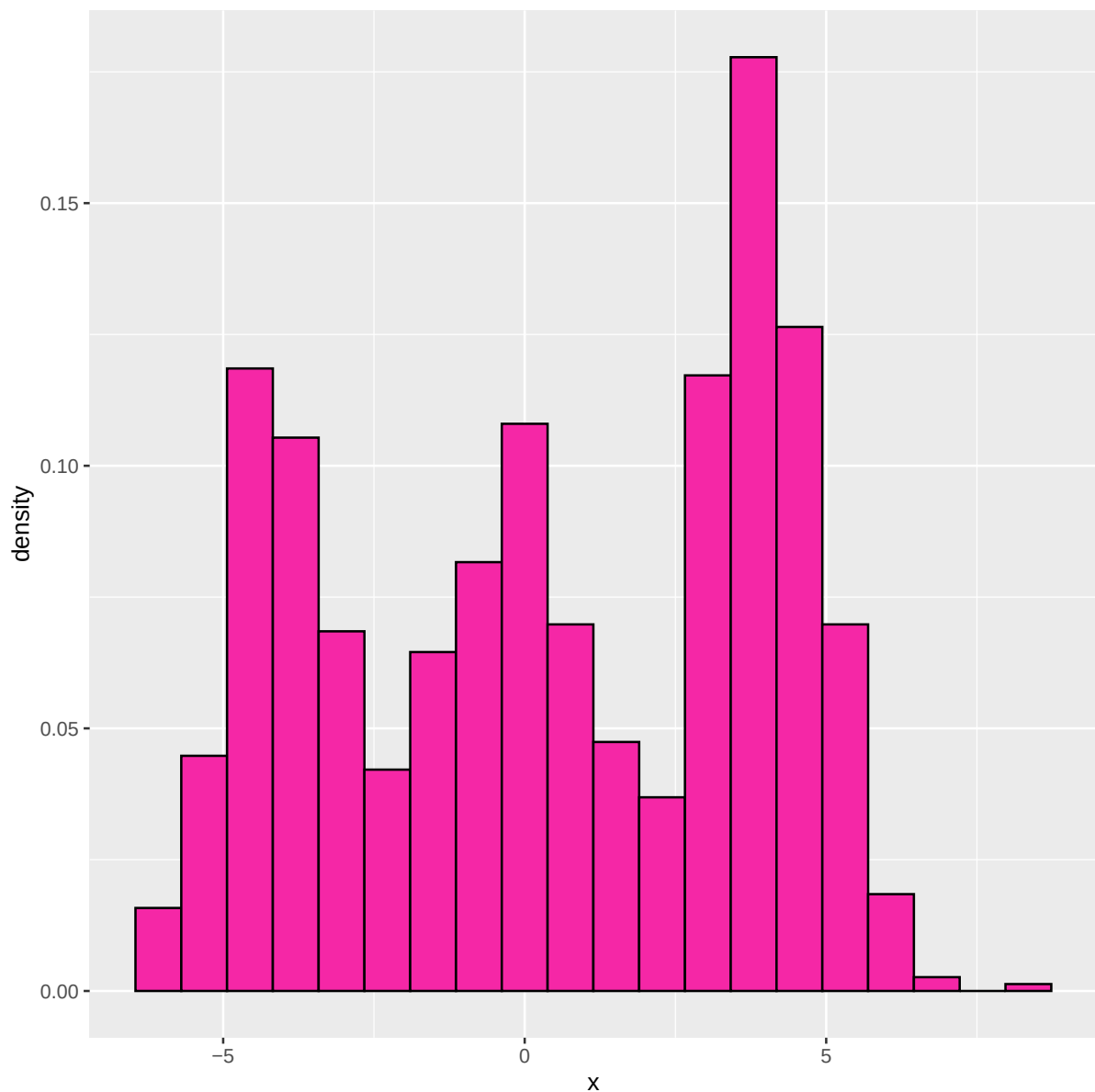
```
z <- sample.int(3, size = n, replace = TRUE, prob = phi.true)
```

```
x <- rnorm(n, mean = mu.true[z], sd = sqrt(sigma.sq.true[z]))
```

Here is a histogram of the generated sample.

```
library(tidyverse)
```

```
data.frame(x) %>%  
  ggplot(aes(x = x)) +  
  geom_histogram(aes(y = after_stat(density)), bins = 20,  
                 color = "black", fill = "#F527A6")
```



 Clearly the sample is from a multimodal distribution, each peak governed by a Gaussian.

```
2 theta <- c(0.4, 0.4, 0.2, -2, 0, 3, 1, 2, 2)
```

```
epsilon <- 0.0001
```

```
repeat{
  # current setting
  phi1 <- theta[1]; phi2 <- theta[2]; phi3 <- theta[3]
  mu1 <- theta[4]; sigma1.sq <- theta[7]
  mu2 <- theta[5]; sigma2.sq <- theta[8]
  mu3 <- theta[6]; sigma3.sq <- theta[9]

  # E-Step
  joint.prob <- matrix(NA, nrow = n, ncol = 3)
  joint.prob[,1] <- dnorm(x, mu1, sqrt(sigma1.sq)) * phi1
  joint.prob[,2] <- dnorm(x, mu2, sqrt(sigma2.sq)) * phi2
  joint.prob[,3] <- dnorm(x, mu3, sqrt(sigma3.sq)) * phi3

  marginal.prob <- rowSums(joint.prob)

  posterior.prob <- joint.prob / marginal.prob # using recycling property of R

  # M-Step

  theta.new <- c()

  # update phi
  theta.new[c(1, 2, 3)] <- colMeans(posterior.prob)

  # update mu
  theta.new[c(4, 5, 6)] <- colSums(posterior.prob * x) / colSums(posterior.prob)

  # update sigma.sq
  temp <- (x - matrix(rep(c(mu1, mu2, mu3), n), nrow = n, ncol = 3, byrow = TRUE))^2

  theta.new[c(7, 8, 9)] <- colSums(posterior.prob * temp) / colSums(posterior.prob)

  if(max(abs(theta - theta.new)) < epsilon){
    theta <- theta.new
    break
  }else{
    theta <- theta.new
  }
}
```

```
theta
```

```
## [1] 0.2739942 0.3118121 0.4141937 -4.1205763 -0.2417299 4.0106135 0.7660326
## [8] 1.4375176 0.9193714
```

3

We now calculate the posterior probabilities using the latest setting of the parameters, hence provide an estimate of which Gaussian each $x^{(i)}$ might belong to.

```
phi1 <- theta[1]; phi2 <- theta[2]; phi3 <- theta[3]
mu1 <- theta[4]; sigma1.sq <- theta[7]
mu2 <- theta[5]; sigma2.sq <- theta[8]
mu3 <- theta[6]; sigma3.sq <- theta[9]
```

```
joint.prob <- matrix(NA, nrow = n, ncol = 3)
joint.prob[,1] <- dnorm(x, mu1, sqrt(sigma1.sq)) * phi1
joint.prob[,2] <- dnorm(x, mu2, sqrt(sigma2.sq)) * phi2
joint.prob[,3] <- dnorm(x, mu3, sqrt(sigma3.sq)) * phi3
```

```
marginal.prob <- rowSums(joint.prob)
```

```
posterior.prob <- joint.prob / marginal.prob
```

```
head(posterior.prob)
```

```
##           [,1]      [,2]      [,3]
## [1,] 1.999264e-03 0.9979976574 3.079112e-06
## [2,] 9.356060e-03 0.9906433829 5.575712e-07
## [3,] 1.896452e-20 0.0006561977 9.993438e-01
## [4,] 2.069683e-03 0.9979273488 2.967862e-06
## [5,] 8.922187e-01 0.1077812578 5.235876e-12
## [6,] 1.733805e-03 0.9982626145 3.580217e-06
```

```
assignment <- data.frame(sample = x,
                        predicted.Gaussian = max.col(posterior.prob))
```

Table 1: Assignment to Gaussians

sample	predicted.Gaussian
-0.948	2
-1.290	2
4.192	3
-0.955	2
-3.030	1
-0.917	2
-4.490	1
-3.829	1
3.861	3
3.567	3

➔ Conclusion

Parameter	Estimate	True Value
$\widehat{\phi}_1$	0.274	0.3
$\widehat{\phi}_2$	0.312	0.3
$\widehat{\phi}_3$	0.414	0.4
$\widehat{\mu}_1$	-4.121	-4
$\widehat{\mu}_2$	-0.242	0
$\widehat{\mu}_3$	4.011	4
$\widehat{\sigma}_1^2$	0.766	1
$\widehat{\sigma}_2^2$	1.438	1
$\widehat{\sigma}_3^2$	0.919	1

Table 2: Estimated and true parameter values

 Even though EM algorithm has a tendency to converge to the local optima, it has performed considerably well in our problem.